HEALTHCARE DATA ANALYSIS USING SQL

SUBMITTED BY:

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GENDER-WISE PATIENT ANALYSIS

Query :

```
select (SELECT count(*) from patients where gender ='Male') as male_count,(select count(*) from patients where gender ='Female') as female_count;
```

Output:

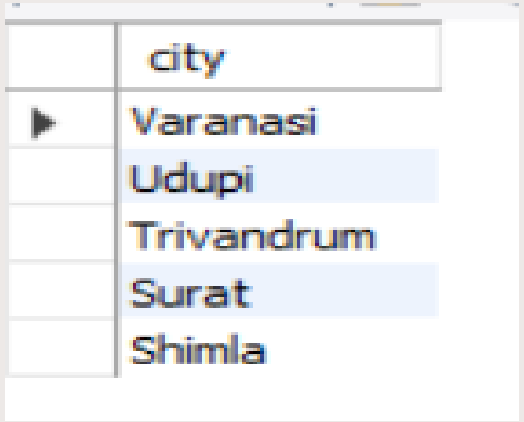
	male_count	female_count
▶	77	73

PATIENT DISTRIBUTION BY CITY

Query :

```
select city from patients group by city order by city desc limit 5;
```

Output :



city
Varanasi
Udupi
Trivandrum
Surat
Shimla

DEPARTMENT PERFORMANCE

Query :

```
select
sum(consultation_fee) as total_consultation_revenue,
department, count(*) as total_appointments
from appointments
group by appointment_id;
```

Sample Output:

total_consultation_revenue	department	total_appointments
800	Cardiology	1
500	Dermatology	1
700	Orthopedics	1
900	Gynecology	1
1200	Neurology	1
400	General	1
600	ENT	1
500	Pediatrics	1

DOCTOR WORKLOAD ANALYSIS

Query :

```
select doctor_name,count(*) as appointments_handeled ,
 Dense_rank() over(order by count(*) desc) as rank_of_doctors
 from appointments group by doctor_name ;
```

Sample Output:

doctor_name	appointments_handeled	rank_of_doctors
Dr. Rao	3	1
Dr. Nair	2	2
Dr. Kapoor	2	2
Dr. Iyer	2	2
Dr. Verma	2	2
Dr. Shetty	2	2
Dr. Arora	2	2
Dr. Das	1	3

REVENUE BY DOCTOR

Query :

```
select doctor_name,sum(consultation_fee) as total_consultation_revenue ,
avg(consultation_fee) as average_consultation_revenue
from appointments
group by doctor_name
order by total_consultation_revenue  desc limit 5;
```

Output :

doctor_name	total_consultation_revenue	average_consultation_revenue
Dr. Verma	2400	1200.0000
Dr. Kapoor	1850	925.0000
Dr. Iyer	1750	875.0000
Dr. Rao	1650	550.0000
Dr. Shetty	1300	650.0000

PATIENTS WITHOUT APPOINTMENTS

Query :

```
select patients.patient_id,appointments.appointment_id
from patients left join appointments on patients.patient_id=appointments.patient_id
where appointment_id is null;
```

Sample Output :

patient_id	appointment_id
51	NULL
52	NULL
53	NULL
54	NULL
55	NULL
56	NULL
57	NULL
58	NULL

APPOINTMENT REVENUE BY MONTH

Query :

```
select year(appointment_date),month(appointment_date),count(*),sum(consultation_fee)
from appointments
group by appointment_date,consultation_fee
order by year(appointment_date),month(appointment_date) asc ;
```

Sample Output:

year(appointment_date)	month(appointment_date)	count(*)	sum(consultation_fee)
2023	2	1	800
2023	2	1	500
2023	2	1	700
2023	2	1	900
2023	3	1	1200
2023	3	1	400
2023	3	1	600
2023	3	1	500

DEPARTMENT FEE COMPARISON

Query :

```
select department, avg(consultation_fee) as avg_fee from appointments group by department  order by avg_fee desc ;
```

Output :

department	avg_fee
Neurology	1180.0000
Gynecology	930.0000
Cardiology	843.7500
Orthopedics	716.6667
ENT	625.0000
Dermatology	531.2500
Pediatrics	512.5000
General	418.7500

FIRST AND LATEST APPOINTMENT

Query:

```
select patients.patient_id,max(appointments.appointment_date) as max_appointment,
min(appointments.appointment_id) as min_appointment
from patients join appointments on patients.patient_id=appointments.patient_id
group by patients.patient_id;
```

Sample output:

patient_id	max_appointment	min_appointment
1	2023-02-10	101
2	2023-02-15	102
3	2023-02-20	103
4	2023-02-25	104
5	2023-03-01	105
6	2023-03-05	106
7	2023-03-10	107
8	2023-03-12	108

RANK DEPARTMENTS BY REVENUE

Query:

```
with department_r as (
select department,sum(consultation_fee) as total_revenue
from appointments
group by department )
select department , total_revenue, dense_rank() over(order by total_revenue desc) as revenue_rank from department_r;
```

Sample output:

department	total_revenue	revenue_rank
Cardiology	6750	1
Neurology	5900	2
Gynecology	4650	3
Orthopedics	4300	4
Dermatology	4250	5
ENT	3750	6
General	3350	7
Pediatrics	2050	8

DATA QUALITY CHECK

Query :

```
select appointments.patient_id,patients.patient_id from appointments
left join patients on appointments.patient_id=patients.patient_id
where patients.patient_id is null;
```

Output:

	patient_id	patient_id

TOP DEPARTMENTS BY MONTH

Query :

```
with department_r as (
select
department,sum(consultation_fee) as total_revenue,
     month(appointment_date) as months
from appointments
group by department,month(appointment_date) )
select department ,months, total_revenue, rank() over(partition by months order by total_revenue desc) as revenue_rank from department_r ;
```

Sample output :

department	months	total_revenue	revenue_rank
Gynecology	2	900	1
Cardiology	2	800	2
Orthopedics	2	700	3
Dermatology	2	500	4
Neurology	3	2300	1

ABOVE-AVERAGE CONSULTATION FEES

Query :

```
select * from appointments where consultation_fee >(select avg(consultation_fee) from appointments);
```

Sample output:

appointment_id	patient_id	appointment_date	doctor_name	department	consultation_fee
101	1	2023-02-10	Dr. Rao	Cardiology	800
104	4	2023-02-25	Dr. Iyer	Gynecology	900
105	5	2023-03-01	Dr. Verma	Neurology	1200
109	9	2023-03-15	Dr. Patel	Cardiology	800
113	13	2023-03-28	Dr. Joshi	Neurology	1100

DEPARTMENT REVENUE CONTRIBUTION

Query :

```
select department,sum(consultation_fee) as total_fee,sum(consultation_fee)*100.0/(select sum(consultation_fee )
from appointments) as percentation_consultation_fee from appointments group by department;
```

Sample output:

department	total_fee	percentation_consultation_fee
Cardiology	6750	19.28571
Dermatology	4250	12.14286
Orthopedics	4300	12.28571
Gynecology	4650	13.28571
Neurology	5900	16.85714

AGE GROUP ANALYSIS

Query:

```
select
case
      when timestampdiff(year,date_of_birth,curdate()) <18
  then 'under 18'
    when timestampdiff(year,date_of_birth,curdate()) between 18 and 30
    then '18-30'
      when timestampdiff(year,date_of_birth,curdate()) between 31 and 45
      then'31-45'
        when timestampdiff(year,date_of_birth,curdate()) between 46 and 60
        then '46-60'
  else 'over 60'
    else 'over 60'
    end as age_categorie,
count(*) as patient_count
from patients group by age_categorie order by min(timestampdiff(year,date_of_birth,curdate())) ;
```

Sample output :

age_categorie	patient_count
18-30	18
31-45	132

THANK YOU